CANDIDATE'S NAME: ______________________________ CTG: _________________

YISHUN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2018

CHEMISTRY **8873/01**
HIGHER 1 **13 September 2018**
Paper 1: Multiple-Choice Questions **1400hrs – 1500hrs**
1 hour

Additional materials:

Optical Mark Sheet
Data Booklet

INSTRUCTIONS TO CANDIDATES

Multiple Choice Questions

Write in soft pencil.

Do not use paper clips, highlighters, glue or correction fluid.

Write your name and CTG on the Answer Sheet. Shade your NRIC number in the spaces provided in the Optical Mark Sheet.

There are thirty questions on this paper. Answer **all** questions. For each question, there are four possible answers **A**, **B**, **C**, and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 Nitrogen and phosphorus are both in Group 15 of the Periodic Table. Phosphorus forms a chloride with the formula PCl_5 but nitrogen does not form NCl_5.

Which statements help to explain this?

1 Nitrogen cannot have an oxidation state of +5.

2 Nitrogen is more electronegative than phosphorus.

3 Nitrogen's outer shell cannot contain more than eight electrons.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 3 only

2 When nuclear reactions take place, the elements produced are different from the elements that reacted. Nuclear equation, such as the one below, are used to represent the changes that occur.

$$^{235}_{92}U + \, ^{1}_{0}n \rightarrow \, ^{144}_{56}Ba + \, ^{89}_{36}Kr + 3\,^{1}_{0}n$$

The nucleon (mass) number total is constant at 236 and the proton number total is constant at 92.

In another nuclear reaction, uranium-238 is reacted with deuterium atoms, $^{2}_{1}H$. An isotope of a new element, **G** is formed as well as two neutrons.

$$^{238}_{92}U + \, ^{2}_{1}H \rightarrow G + 2\,^{1}_{0}n$$

What is isotope **G**?

A ^{238}Np **B** ^{238}Pu **C** ^{240}Np **D** ^{240}Pu

3 Two conversions are shown.

$$NH_4^+ \rightarrow NH_3$$
$$C_2H_4 \rightarrow C_2H_6$$

Which similar features do these conversions have?

A change in oxidation state of an element

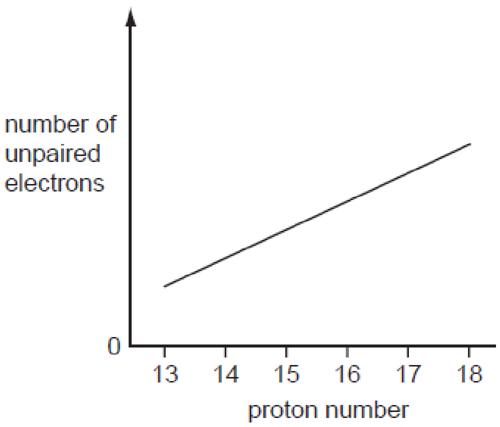
B decrease in bond angle

C formation of a lone pair of electrons

D loss of a π bond

4 Which graph represents the number of unpaired p orbital electrons for atoms with proton number 13 to 18?

A



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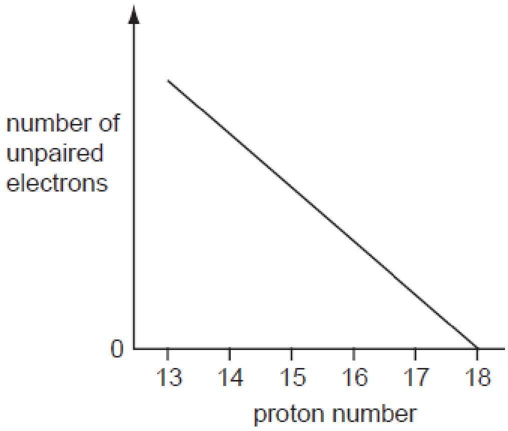

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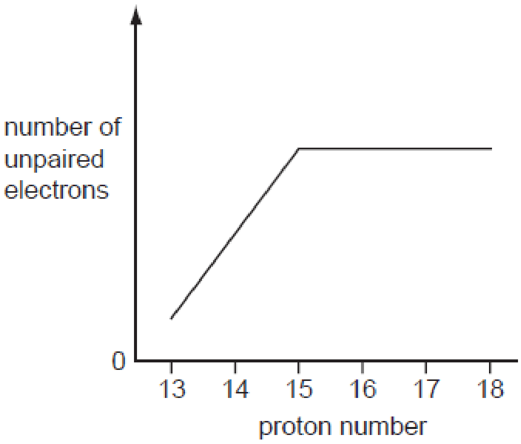

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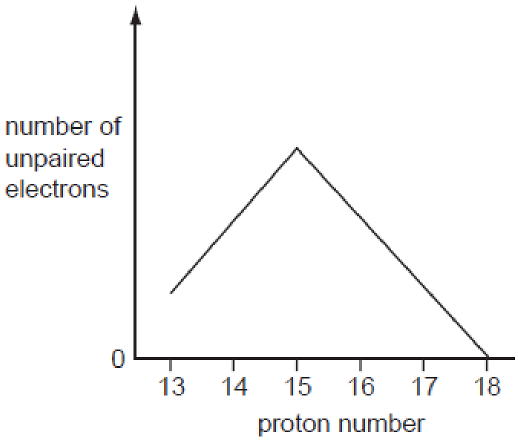


5 What is the correct number of bonds of each type in the Al_2Cl_6 molecule?

	covalent	co-ordinate (dative covalent)
A	6	1
B	6	2
C	7	0
D	7	1

6 Ethylene glycol, $HOCH_2CH_2OH$, is primarily used in antifreeze formulations and as a raw material in the manufacture of polyesters. It has a boiling point of 197 °C while ethanol boils at 79 °C.

Which statements help account for this difference in melting points?

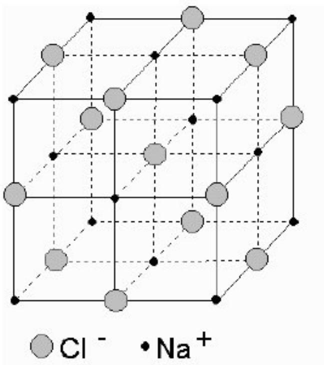
1 Ethylene glycol forms more hydrogen bonds per molecule than ethanol.

2 Ethylene glycol has a larger surface area than ethanol.

3 The covalent bonds in ethylene glycol are stronger than those in ethanol.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1 only

7 In the sodium chloride lattice the number of chloride ions that surround each sodium ion is called the *co-ordination number* of the sodium ions.



Referring to the above diagram, what are the co-ordination numbers of the sodium ions and the chloride ions in the sodium chloride lattice?

	sodium ions	chloride ions
A	4	4
B	6	4
C	6	6
D	4	6

8 The pH of human blood is constant at about 7.40.

Which ion or molecule in the human body will remove contaminating $H^+(aq)$ ions from the blood to keep the pH constant?

A CO_3^{2-} **B** HCO_3^- **C** H_2CO_3 **D** PO_4^{3-}

9 What is the final pH of the solution formed by mixing equal volumes of two NaOH solutions, one with pH 11.0 and the other with pH 13.0?

A 11.7 **B** 12.0 **C** 12.7 **D** 13.0

10 **J**, **K** and **L** are three elements in the third period.

- **J** reacts with chlorine to give a liquid product.
- **K** reacts with chlorine to give a solid product that dissolves in water to give a solution of pH 7.
- **L** reacts with chlorine to give a solid product that dissolves in water to give a solution of pH 6.

Which elements are good conductors of electricity?

A **J** only **B** **J** and **K** **C** **K** only **D** **K** and **L**

11 The volatility of the Group 17 elements chlorine, bromine, and iodine, decreases down the group.

What is responsible for this?

A bond length in the halogen molecule

B bond strength in the halogen molecule

C electronegativity of the halogen

D number of electrons in the halogen molecule

12 This question refers to isolated gaseous species.

The species F^-, Ne and Na^+ are isoelectronic. This means that they have the same number of electrons.

In which order do their radii increase?

	smallest	$\longrightarrow$	largest
A	Na^+	Ne	F^-
B	Na^+	F^-	Ne
C	F^-	Ne	Na^+
D	F^-	Na^+	Ne

13 A student makes sodium chloride by reacting 0.0125 mol of sodium carbonate with an excess of 0.50 mol dm^{-3} hydrochloric acid.

$$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$$

Which statements about the quantities of substance are correct?

1 300 cm^3 of carbon dioxide is produced at room temperature and pressure.

2 25 cm^3 of the hydrochloric acid is needed to exactly neutralise the sodium carbonate.

3 1.46 g of sodium chloride is produced.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

14 Shakudo is a Japanese alloy of copper and gold. The information in the table as obtained by mass spectrometry of a sample of shakudo.

mass number	63	65	197
% abundance	65	29	6

What is the A_r value for copper?

A 59.8 **B** 63.5 **C** 63.6 **D** 71.6

15 $Na_2S_2O_3$ reacts with HC*l* as shown.

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l) + SO_2(g) + S(s)$$

What is the oxidation number of sulfur in $Na_2S_2O_3$, SO_2 and S?

	$Na_2S_2O_3$	SO_2	S
A	+2	+2	+1
B	+2	+4	0
C	+4	+2	+1
D	+4	+4	0

16 Given the following half equations:

$$Cr_2O_7^{2-} + 14\,H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$$

$$Fe^{2+} \rightarrow Fe^{3+} + e^-$$

$$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$$

What volume of 0.020 mol dm^{-3} of potassium dichromate, $K_2Cr_2O_7$, is required to oxidise 18 cm^3 of an acidified solution of 0.020 mol dm^{-3} of FeC_2O_4 ?

A 3.0 cm^3 **B** 6.0 cm^3 **C** 9.0 cm^3 **D** 18.0 cm^3

17 In a calorimetric experiment 1.60 g of a fuel are burnt. 45.0 % of the energy released is absorbed by 200 g of water. The temperature of the water rises from 18.0 °C to 66.0 °C.

Given that the specific heat capacity of water is 4.18 J g^{-1} K^{-1}, what is the total energy released per gram of fuel burnt (to 3 significant figures)?

A 11 300 J **B** 55 700 J **C** 89 200 J **D** 143 000 J

18 Which equation represents the standard enthalpy change of formation of water?

A $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$

B $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$

C $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$

D $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

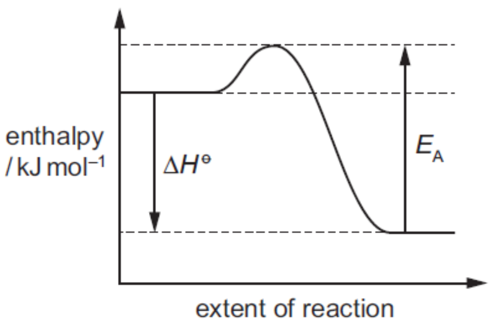
19 Nitric oxide, NO and bromine vapour react together according to the following equation.

$$2NO(g) + Br_2(g) \rightarrow 2NOBr\ (g) \qquad \Delta H^{\ominus} = -23\ \text{kJ mol}^{-1}$$

The reaction has an activation energy of +5.4 kJ mol^{-1}.

What is the correct reaction pathway diagram for this reaction?

A



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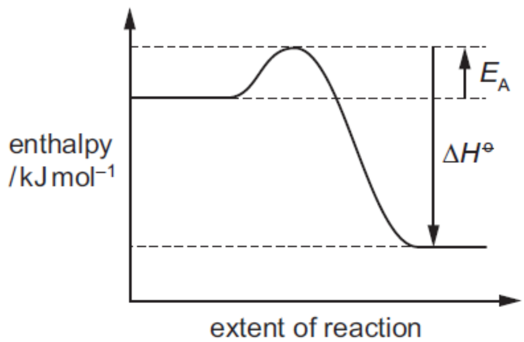

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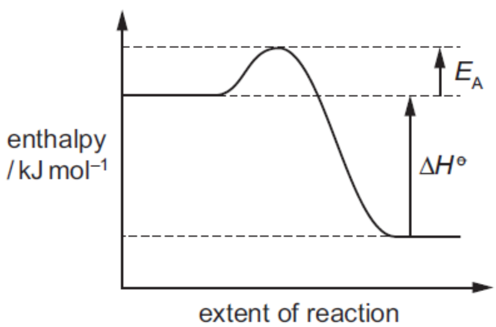

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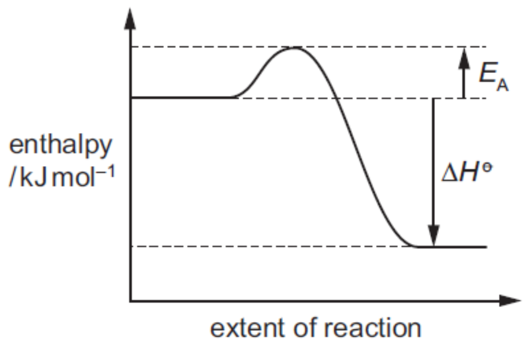


20 Enzymes are biological catalysts. Many enzymes show specificity. An example of an enzyme which shows specificity is glucokinase. Glucokinase is involved in the metabolism of glucose.

What does specificity mean in this context?

A Glucokinase is most effective as a catalyst over a narrow pH range.

B Glucokinase is most effective as a catalyst over a narrow range of temperatures.

C Glucokinase only operates on a narrow range of substrate molecules.

D Glucokinase provides an alternative route for the reactions it catalyses.

21 Methanol, CH_3OH, can be produced industrially by reacting CO with H_2.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H = -91 \text{ kJ mol}^{-1}$$

The process can be carried out at 4 x 10^3 kPa and 1150 K.

Which statements about the reaction are correct?

1 Increasing the temperature will increase the rate of the backward reaction.

2 Increasing the temperature will decrease the yield of methanol.

3 Increasing the pressure will increase the yield of methanol.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

22 Two moles of **P** were placed in a sealed container. The container was heated and **P** was partially decomposed to produce **Q** and **R** only. A dynamic equilibrium between **P**, **Q** and **R** was established.

At equilibrium, *x* moles of **R** were present and the total number of moles present was $\left(2+\frac{x}{2}\right)$.

What is the equation for this reversible reaction?

A **P** ⇌ 2**Q** + **R**

B 2**P** ⇌ 2**Q** + **R**

C 2**P** ⇌ **Q** + **R**

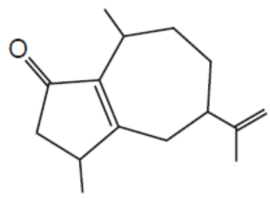
D 2**P** ⇌ **Q** + 2**R**

23 Researchers have attempted to create a synthetic material that sticks to surfaces, by imitating the mechanism that allows the gecko to climb a vertical surface.

Which statement explains how the synthetic material sticks to surfaces?

A It forms a glue that forms covalent bonds with surfaces.
B It forms a glue that forms van der Waals' forces with surfaces.
C It is a nanomaterial that has a high surface area to volume ratio.
D It is a nanomaterial that has a low surface area to volume ratio.

24 The compound rotundone is responsible for the peppery smell of pepper and is also found in some red wines.



rotundone

How many hydrogen atoms are in one molecule of rotundone?

A 15	**B** 19	**C** 22	**D** 24

25 *Melamine* and *acrylic* are both hard polymers. *Melamine* is used on worktop surfaces and it chars (becomes blackened) on heating. *Acrylic* is a sturdy substitute for glass and it melts on heating.

Which statements about *melamine* and *acrylic* are correct?

1 Both polymers can be recycled.
2 Heating does not break the crosslinks in *acrylic*.
3 Heating does not break the crosslinks in *Melamine.*

A 3 only	**B** 1 and 2 only	**C** 1 and 3 only	**D** 2 and 3 only

26 It is difficult to dispose PVC. Two possible methods are burying it in landfill sites and disposal by combustion.

Which option is correct?

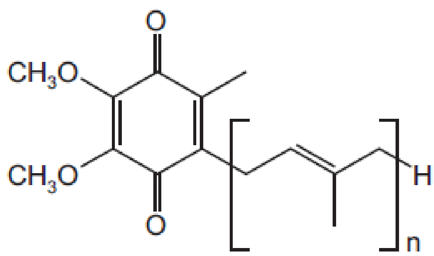
	rate of biodegradation of PVC in landfill sites	gases produced when PVC combusts
A	fast	CO_2, H_2O, HCl
B	fast	CO_2, H_2O, Cl_2
C	slow	CO_2, H_2O, HCl
D	slow	CO_2, H_2O, Cl_2

27 How many esters with the molecular formula $C_5H_{10}O_2$ can be made by reacting a **primary** alcohol with a carboxylic acid?

A 4 **B** 5 **C** 6 **D** 8

28 People who take statin drugs to control their blood cholesterol may also take 'coenzyme Q_{10}'.

The diagram shows a simplified structure of one form of this coenzyme.

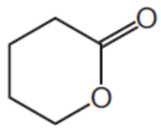


coenzyme Q_{10}

Which option describes this structure correctly?

	functional groups present	number of π bonds in one molecule
A	aldehyde and alkene	n + 2
B	aldehyde and alkene	n + 4
C	ketone and alkene	n + 2
D	ketone and alkene	n + 4

29 Cyclic esters are also known as lactones. *Delta* lactone is used as solvent and in the manufacture of polyesters.



delta lactone

Which compound could be used to produce *delta* lactone in a single reaction?

A $HOCH_2CH_2CH_2COCH_3$

B $HOCH_2CH_2CH_2CH_2CHO$

C $HOCH_2CH_2CH_2CH_2CO_2H$

D $HOCH_2CH_2CH_2CH_2CH_2OH$

30 Compound **X** is heated with a mild oxidising agent. One of the products of the reaction will react with hydrogen cyanide, forming 2-hydroxypentanenitrile.

What is compound **X**?

A butan-1-ol
B butan-2-ol
C pentan-1-ol
D pentan-2-ol